

## Reverse-check review package — topBSM\_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

| item                | value                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model file          | topBSM/model/topBSM_gen.fr                                                                                                                                                         |
| original model name | topBSM_gen (hidden from the agent)                                                                                                                                                 |
| paper               | topBSM/text/1305.7386.txt                                                                                                                                                          |
| blindness scope     | blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back |

### Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`bjzeros (=)`

```
Table[N[BesselJZero[1, x]], {x, 1, 10}]
```

```
M$Parameters = {
  s0scalar == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> SOPARAMS,
    InteractionOrder -> {QS0, 1},
    ParameterName -> s0scalar,
    TeX -> "Subscript[s0, scalar]",
    Description -> "S0 scalar coupling"
  },
  s0axial == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> SOPARAMS,
    InteractionOrder -> {QS0, 1},
    ParameterName -> s0axial,
    TeX -> "Subscript[s0, axial]",
    Description -> "S0 axial coupling"
  },
  s0fusionScalar == {
    ParameterType -> Internal,
    InteractionOrder -> {QS0, 1},
    Definitions -> {s0fusionScalar :> If[NumericalValue[MS0] > 2 NumericalValue[MT],
    ↪ -s0scalar gs^2/(12 Pi^2 vev) sertHeavy[(2 MT/MS0)^2], -s0scalar gs^2/(12 Pi^2 vev)
    ↪ sertLight[(2 MT/MS0)^2]]},
    TeX -> "Subscript[s0, fusionScalar]",
    Description -> "S0 effective coupling due to gluon fusion, scalar"
  },
  s0fusionAxial == {
    ParameterType -> Internal,
    InteractionOrder -> {QS0, 1},
```

```

Definitions -> {sOfusionAxial :> If[NumericalValue[MS0] > 2 NumericalValue[MT], -s0axial
↵ gs^2/(8 Pi^2 vev) serpHeavy[(2 MT/MS0)^2], -s0axial gs^2/(8 Pi^2 vev) serpLight[(2
↵ MT/MS0)^2]]},
TeX -> "Subscript[s0, fusionAxial]",
Description -> "S0 effective coupling due to gluon fusion, axial"
},
WS0 == {
ParameterType -> Internal,
Definitions -> {WS0 :> If[NumericalValue[MS0] > 2 NumericalValue[MT], (3 MT^2 Sqrt[MS0^4
↵ - 4 MS0^2 MT^2] (-4 MT^2 sOscalar^2 + MS0^2 (s0axial^2 + sOscalar^2)))/(8 Pi vev^2 MS0^3)
↵ + (MS0^3 Abs[sOfusionScalar]^2)/(8 Pi), (MS0^3 Abs[sOfusionScalar]^2)/(8 Pi)]},
Description -> "S0 width"
},
oOscalar == {
ParameterType -> External,
Value -> 1,
BlockName -> OOPARAMS,
InteractionOrder -> {Q00, 1},
ParameterName -> oOscalar,
TeX -> "Subscript[o0, scalar]",
Description -> "00 scalar coupling"
},
oOaxial == {
ParameterType -> External,
Value -> 1,
BlockName -> OOPARAMS,
InteractionOrder -> {Q00, 1},
ParameterName -> oOaxial,
TeX -> "Subscript[o0, axial]",
Description -> "00 axial coupling"
},
oOfusionScalar == {
ParameterType -> Internal,
InteractionOrder -> {Q00, 1},
Definitions -> {oOfusionScalar :> If[NumericalValue[M00] > 2 NumericalValue[MT],
↵ -oOscalar gs^2/(12 Pi^2 vev) sertHeavy[(2 MT/M00)^2], -oOscalar gs^2/(12 Pi^2 vev)
↵ sertLight[(2 MT/M00)^2]]},
TeX -> "Subscript[o0, fusionScalar]",
Description -> "00 effective coupling due to gluon fusion, scalar"
},
oOfusionAxial == {
ParameterType -> Internal,
InteractionOrder -> {Q00, 1},
Definitions -> {oOfusionAxial :> If[NumericalValue[M00] > 2 NumericalValue[MT], -oOaxial
↵ gs^2/(8 Pi^2 vev) serpHeavy[(2 MT/M00)^2], -oOaxial gs^2/(8 Pi^2 vev) serpLight[(2
↵ MT/M00)^2]]},
TeX -> "Subscript[o0, fusionAxial]",
Description -> "00 effective coupling due to gluon fusion, axial"
},
W00 == {
ParameterType -> Internal,
Definitions -> {W00 :> If[NumericalValue[M00] > 2 NumericalValue[MT], 1/6 (3 MT^2
↵ Sqrt[M00^4 - 4 M00^2 MT^2] (-4 MT^2 oOscalar^2 + M00^2 (oOaxial^2 + oOscalar^2)))/(8 Pi
↵ vev^2 M00^3) + 1/64 (M00^3 Abs[oOfusionScalar]^2)/(8 Pi), 1/64 (M00^3
↵ Abs[oOfusionScalar]^2)/(8 Pi)]},

```

```

    Description -> "00 width"
  },
  sluright == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> S1PARAMS,
    InteractionOrder -> {QS1, 1},
    ParameterName -> sluright,
    TeX -> "Subscript[s1, ur]",
    Description -> "S1 right up quark coupling"
  },
  sluleft == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> S1PARAMS,
    InteractionOrder -> {QS1, 1},
    ParameterName -> sluleft,
    TeX -> "Subscript[s1, ul]",
    Description -> "S1 left up quark coupling"
  },
  sidright == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> S1PARAMS,
    InteractionOrder -> {QS1, 1},
    ParameterName -> sidright,
    TeX -> "Subscript[s1, dr]",
    Description -> "S1 right down quark coupling"
  },
  sidleft == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> S1PARAMS,
    InteractionOrder -> {QS1, 1},
    ParameterName -> sidleft,
    TeX -> "Subscript[s1, dl]",
    Description -> "S1 left down quark coupling"
  },
  sleright == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> S1PARAMS,
    InteractionOrder -> {QS1, 1},
    ParameterName -> sleright,
    TeX -> "Subscript[s1, er]",
    Description -> "S1 right electron coupling"
  },
  sleleft == {
    ParameterType -> External,
    Value -> 1,
    BlockName -> S1PARAMS,
    InteractionOrder -> {QS1, 1},
    ParameterName -> sleleft,
    TeX -> "Subscript[s1, el]",
    Description -> "S1 left electron coupling"
  }

```

```

},
sinu == {
  ParameterType -> External,
  Value -> 1,
  BlockName -> S1PARAMS,
  InteractionOrder -> {QS1, 1},
  ParameterName -> sinu,
  TeX -> "Subscript[s1, v]",
  Description -> "S1 neutrino coupling"
},
WS1 == {
  ParameterType -> Internal,
  Definitions -> {WS1 -> 1/(288*cw^2*Pi*sw^2*MS1^3)*ee^2*(+9*MS1^4*sinu^2
↵ +2*MS1^4*(16*sluright^2*sw^4 + sluleft^2*(3 - 4*sw^2)^2) +6*MS1^4*(4*sleright^2*sw^4 +
↵ slieleft^2*(1 - 2*sw^2)^2) +2*MS1^4*(4*sidright^2*sw^4 + sidleft^2*(3 - 2*sw^2)^2)
↵ +Sqrt[MS1^4 - 4*MS1^2*MT^2]*(+MT^2*(-9*sluleft^2 +24*sluleft*(sluleft - 3*sluright)*sw^2
↵ -16*(sluleft^2 - 6*sluleft*sluright + sluright^2)*sw^4) +MS1^2*(16*sluright^2*sw^4 +
↵ sluleft^2*(3 - 4*sw^2)^2)) +3*Sqrt[MS1^4 - 4*MS1^2*MTA^2]*(-MTA^2*(+slieleft^2
↵ -4*slieleft*(slieleft - 3*sleright)*sw^2 +4*(slieleft^2 - 6*slieleft*sleright +
↵ sleright^2)*sw^4) +MS1^2*(4*sleright^2*sw^4 + slieleft^2*(1 - 2*sw^2)^2))
↵ +Sqrt[-4*MB^2*MS1^2 + MS1^4]*(+MB^2*(-9*sidleft^2 +12*sidleft*(sidleft -
↵ 3*sidright)*sw^2 -4*(sidleft^2 - 6*sidleft*sidright + sidright^2)*sw^4)
↵ +MS1^2*(4*sidright^2*sw^4 + sidleft^2*(3 - 2*sw^2)^2)))},
  Description -> "S1 width"
},
oluright == {
  ParameterType -> External,
  Value -> 1,
  BlockName -> O1PARAMS,
  InteractionOrder -> {QO1, 1},
  ParameterName -> oluright,
  TeX -> "Subscript[o1, ur]",
  Description -> "O1 right up quark coupling"
},
oluleft == {
  ParameterType -> External,
  Value -> 1,
  BlockName -> O1PARAMS,
  InteractionOrder -> {QO1, 1},
  ParameterName -> oluleft,
  TeX -> "Subscript[o1, ul]",
  Description -> "O1 left up quark coupling"
},
oldright == {
  ParameterType -> External,
  Value -> 1,
  BlockName -> O1PARAMS,
  InteractionOrder -> {QO1, 1},
  ParameterName -> oldright,
  TeX -> "Subscript[o1, dr]",
  Description -> "O1 right down quark coupling"
},
oldleft == {
  ParameterType -> External,
  Value -> 1,

```

```

BlockName -> O1PARAMS,
InteractionOrder -> {Q01, 1},
ParameterName -> o1dleft,
TeX -> "Subscript[o1, dl]",
Description -> "O1 left down quark coupling"
},
W01 == {
  ParameterType -> Internal,
  Definitions -> {W01 -> (gs^2/(48*Pi*M01^3))*(+2*M01^4*(o1dleft^2 + o1dright^2)
↵ +Sqrt[-4*MB^2*M01^2 + M01^4]*(M01^2*(o1dleft^2 + o1dright^2) -MB^2*(o1dleft^2 -
↵ 6*o1dleft*o1dright + o1dright^2)) +2*M01^4*(o1uleft^2 + o1uright^2) +Sqrt[M01^4 -
↵ 4*M01^2*MT^2]*(M01^2*(o1uleft^2 + o1uright^2) -MT^2*(o1uleft^2 - 6*o1uleft*o1uright +
↵ o1uright^2)))},
  Description -> "O1 width"
}
}

```

LS0top (=)

s0scalar MT/vev S0 tbar.t + I s0axial MT/vev S0 tbar.Ga[5].t

LS0ggfusionScalar (=)

-1/4 s0fusionScalar S0 FS[G,mu,nu,aa] FS[G,mu,nu,aa]

LS0ggfusionAxial (=)

-1/4 s0fusionAxial S0 FS[G,mu,nu,aa] Dual[FS][G,mu,nu,aa]

LS0ggfusion (=)

LS0ggfusionScalar + LS0ggfusionAxial

LS0 (=)

LS0top + LS0ggfusion

L00top (=)

o0scalar MT/vev tbar.T[a].t O0[a] + I o0axial MT/vev tbar.Ga[5].T[a].t O0[a]

L00ggfusionScalar (=)

-1/4 o0fusionScalar dSUN[aa,bb,cc] O0[aa] FS[G,mu,nu,bb] FS[G,mu,nu,cc]

L00ggfusionAxial (=)

-1/4 o0fusionAxial dSUN[aa,bb,cc] O0[aa] FS[G,mu,nu,bb] Dual[FS][G,mu,nu,cc]

L00ggfusion (=)

L00ggfusionScalar + L00ggfusionAxial

L00 (=)

L00top + L00ggfusion

```

ez (=)
ee/(sw cw)

ey (=)
ee sw/cw

LS1ul (=)
- siuleft S1[mu] QLbar[sp1,1,ff,cc] Ga[mu,sp1,sp2] QL[sp2,1,ff,cc] * ez (-1/2 + sw^2 2/3)

LS1dl (=)
- sidleft S1[mu] QLbar[sp1,2,ff,cc] Ga[mu,sp1,sp2] QL[sp2,2,ff,cc] * ez (1/2 - sw^2/3)

LS1ur (=)
- siuright S1[mu] uRbar.Ga[mu].uR * ey*2/3

LS1dr (=)
- sidright S1[mu] dRbar.Ga[mu].dR * (-ey)/3

LS1el (=)
- sieleft S1[mu] LLbar[sp1,2,ff] Ga[mu,sp1,sp2] LL[sp2,2,ff] * ez (1/2 - sw^2)

LS1nu (=)
- sinu S1[mu] LLbar[sp1,1,ff] Ga[mu,sp1,sp2] LL[sp2,1,ff] * (-ez)/2

LS1er (=)
- sieright S1[mu] lRbar.Ga[mu].lR * (-ey)

LS1 (=)
LS1ul + LS1dl + LS1ur + LS1dr + LS1el + LS1er + LS1nu

L01ul (=)
oiuleft gs O1[mu, a] QLbar[sp1,1,ff,cc1] Ga[mu,sp1,sp2] T[a,cc1,cc2] QL[sp2,1,ff,cc2]

L01dl (=)
oidleft gs O1[mu, a] QLbar[sp1,2,ff,cc1] Ga[mu,sp1,sp2] T[a,cc1,cc2] QL[sp2,2,ff,cc2]

L01ur (=)
oiuright gs O1[mu, a] uRbar.Ga[mu].T[a].uR

L01dr (=)
oidright gs O1[mu, a] dRbar.Ga[mu].T[a].dR

L01 (=)
L01ul + L01dl + L01ur + L01dr

```

## Blank-slate reconstruction

## Reconstructed Physics from `sanitized.fr`

### Lagrangian

Conventions:  $G_{\mu\nu}^a \equiv \text{FS}[G, \mu, \nu, a]$ ,  $\tilde{G}^{a\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}G_{\rho\sigma}^a$ ,  $T^a$  are fundamental  $SU(3)_c$  generators, and  $d^{abc} \equiv \text{dSUN}[a, b, c]$ . Flavor indices are summed where present. The file defines

$$g_Z \equiv \frac{e}{s_W c_W}, \quad e t_W \equiv \frac{e s_W}{c_W}.$$

No `DC[...]` covariant derivatives appear explicitly in the interaction Lagrangian. From the declared gauge quantum numbers,  $S0, S1, S2$  are color and electroweak singlets with  $D_\mu = \partial_\mu$ .  $O0^a, O1_\mu^a$  are color adjoints with  $Y = Q = 0$ , so if their kinetic terms are supplied elsewhere the QCD covariant derivative is

$$(D_\mu X)^a = \partial_\mu X^a + g_s f^{abc} G_\mu^b X^c,$$

with no  $SU(2)_L$  or  $U(1)_Y$  gauge piece.

### LS0top

$$\mathcal{L}_{\text{S0top}} = s0scalar \frac{m_t}{v} S0 \bar{t} t + i s0axial \frac{m_t}{v} S0 \bar{t} \gamma^5 t.$$

### LS0ggfusionScalar

$$\mathcal{L}_{\text{S0ggfusionScalar}} = -\frac{1}{4} s0fusionScalar S0 G_{\mu\nu}^a G^{a\mu\nu}.$$

### LS0ggfusionAxial

$$\mathcal{L}_{\text{S0ggfusionAxial}} = -\frac{1}{4} s0fusionAxial S0 G_{\mu\nu}^a \tilde{G}^{a\mu\nu}.$$

### LS0ggfusion

$$\mathcal{L}_{\text{S0ggfusion}} = \mathcal{L}_{\text{S0ggfusionScalar}} + \mathcal{L}_{\text{S0ggfusionAxial}}.$$

### LS0

$$\mathcal{L}_{\text{S0}} = \mathcal{L}_{\text{S0top}} + \mathcal{L}_{\text{S0ggfusion}}.$$

### L00top

$$\mathcal{L}_{\text{O0top}} = o0scalar \frac{m_t}{v} O0^a \bar{t} T^a t + i o0axial \frac{m_t}{v} O0^a \bar{t} \gamma^5 T^a t.$$

### L00ggfusionScalar

$$\mathcal{L}_{\text{O0ggfusionScalar}} = -\frac{1}{4} o0fusionScalar d^{abc} O0^a G_{\mu\nu}^b G^{c\mu\nu}.$$

### L00ggfusionAxial

$$\mathcal{L}_{\text{O0ggfusionAxial}} = -\frac{1}{4} o0fusionAxial d^{abc} O0^a G_{\mu\nu}^b \tilde{G}^{c\mu\nu}.$$

### L00ggfusion

$$\mathcal{L}_{\text{O0ggfusion}} = \mathcal{L}_{\text{O0ggfusionScalar}} + \mathcal{L}_{\text{O0ggfusionAxial}}.$$

L00

$$\mathcal{L}_{O0} = \mathcal{L}_{O0\text{top}} + \mathcal{L}_{O0\text{ggfusion}}.$$

LS1ul

$$\mathcal{L}_{S1ul} = -s1uleft g_Z \left( -\frac{1}{2} + \frac{2}{3}s_W^2 \right) S1_\mu \bar{u}_{Lfc} \gamma^\mu u_L^{fc}.$$

LS1dl

$$\mathcal{L}_{S1dl} = -s1dleft g_Z \left( \frac{1}{2} - \frac{1}{3}s_W^2 \right) S1_\mu \bar{d}_{Lfc} \gamma^\mu d_L^{fc}.$$

LS1ur

$$\mathcal{L}_{S1ur} = -s1uright \frac{2}{3} et_W S1_\mu \bar{u}_{Rfc} \gamma^\mu u_R^{fc}.$$

LS1dr

$$\mathcal{L}_{S1dr} = +s1dright \frac{1}{3} et_W S1_\mu \bar{d}_{Rfc} \gamma^\mu d_R^{fc}.$$

LS1el

$$\mathcal{L}_{S1el} = -s1eleft g_Z \left( \frac{1}{2} - s_W^2 \right) S1_\mu \bar{e}_{Lf} \gamma^\mu e_L^f.$$

LS1nu

$$\mathcal{L}_{S1nu} = +s1nu \frac{g_Z}{2} S1_\mu \bar{\nu}_{Lf} \gamma^\mu \nu_L^f.$$

LS1er

$$\mathcal{L}_{S1er} = +s1eright et_W S1_\mu \bar{e}_{Rf} \gamma^\mu e_R^f.$$

LS1

$$\mathcal{L}_{S1} = \mathcal{L}_{S1ul} + \mathcal{L}_{S1dl} + \mathcal{L}_{S1ur} + \mathcal{L}_{S1dr} + \mathcal{L}_{S1el} + \mathcal{L}_{S1er} + \mathcal{L}_{S1nu}.$$

L01ul

$$\mathcal{L}_{O1ul} = o1uleft g_s O1_\mu^a \bar{u}_{Lfc_1} \gamma^\mu T_{c_1c_2}^a u_L^{fc_2}.$$

L01dl

$$\mathcal{L}_{O1dl} = o1dleft g_s O1_\mu^a \bar{d}_{Lfc_1} \gamma^\mu T_{c_1c_2}^a d_L^{fc_2}.$$

L01ur

$$\mathcal{L}_{O1ur} = o1uright g_s O1_\mu^a \bar{u}_{Rfc_1} \gamma^\mu T_{c_1c_2}^a u_R^{fc_2}.$$

L01dr

$$\mathcal{L}_{O1dr} = o1dright g_s O1_\mu^a \bar{d}_{Rfc_1} \gamma^\mu T_{c_1c_2}^a d_R^{fc_2}.$$

L01

$$\mathcal{L}_{O1} = \mathcal{L}_{O1ul} + \mathcal{L}_{O1dl} + \mathcal{L}_{O1ur} + \mathcal{L}_{O1dr}.$$



## Field table

| .fr class | Symbol   | Spin | $SU(3)_c$ rep              | $SU(2)_L$ rep                       | $U(1)$ charge<br>/ hyper-charge | Self-conjugate | Mass       |
|-----------|----------|------|----------------------------|-------------------------------------|---------------------------------|----------------|------------|
| S[21]     | S0       | 0    | singlet                    | not declared,<br>singlet by indices | $Q = 0, Y = 0$                  | yes            | MS0 = 400  |
| S[22]     | 00[a]    | 0    | adjoint,<br>index<br>Gluon | not declared,<br>singlet by indices | $Q = 0, Y = 0$                  | yes            | M00 = 400  |
| V[7]      | S1[mu]   | 1    | singlet                    | not declared,<br>singlet by indices | $Q = 0, Y = 0$                  | yes            | MS1 = 2000 |
| V[8]      | 01[mu,a] | 1    | adjoint,<br>index<br>Gluon | not declared,<br>singlet by indices | $Q = 0, Y = 0$                  | yes            | M01 = 2000 |
| T[1]      | S2       | 2    | singlet                    | not declared,<br>singlet by indices | $Q = 0, Y = 0$                  | yes            | MS2 = 500  |

## Parameters

| External parameter | Value | Multiplies                                                     | Physical meaning                                                                                                                |
|--------------------|-------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| s0scalar           | 1     | $S0\bar{t}t$ , and internally<br>s0fusionScalar                | CP-even scalar coupling of color-singlet spin-0 state S0 to top quarks; also controls loop-induced scalar gluon-fusion operator |
| s0axial            | 1     | $iS0\bar{t}\gamma^5 t$ , and internally<br>s0fusionAxial       | CP-odd pseudoscalar coupling of S0 to top quarks; also controls loop-induced dual-gluon operator                                |
| o0scalar           | 1     | $O0^a\bar{T}^a t$ , and internally<br>o0fusionScalar           | CP-even scalar coupling of color-octet spin-0 state 00 to top quarks; also controls loop-induced $d^{abc}O0^a G^b G^c$ operator |
| o0axial            | 1     | $iO0^a\bar{t}\gamma^5 T^a t$ , and internally<br>o0fusionAxial | CP-odd pseudoscalar coupling of 00 to top quarks; also controls loop-induced $d^{abc}O0^a G^b \tilde{G}^c$ operator             |

| External parameter    | Value | Multiplies         | Physical meaning                                                              |
|-----------------------|-------|--------------------|-------------------------------------------------------------------------------|
| <code>s1uright</code> | 1     | <code>LS1ur</code> | Right-handed up-type quark coupling of neutral color-singlet vector <b>S1</b> |
| <code>s1uleft</code>  | 1     | <code>LS1ul</code> | Left-handed up-type quark coupling of <b>S1</b>                               |
| <code>s1dright</code> | 1     | <code>LS1dr</code> | Right-handed down-type quark coupling of <b>S1</b>                            |
| <code>s1dleft</code>  | 1     | <code>LS1dl</code> | Left-handed down-type quark coupling of <b>S1</b>                             |
| <code>s1eright</code> | 1     | <code>LS1er</code> | Right-handed charged-lepton coupling of <b>S1</b>                             |
| <code>s1eleft</code>  | 1     | <code>LS1el</code> | Left-handed charged-lepton coupling of <b>S1</b>                              |
| <code>s1nu</code>     | 1     | <code>LS1nu</code> | Left-handed neutrino coupling of <b>S1</b>                                    |
| <code>o1uright</code> | 1     | <code>L01ur</code> | Right-handed up-type quark coupling of color-octet vector <b>O1</b>           |
| <code>o1uleft</code>  | 1     | <code>L01ul</code> | Left-handed up-type quark coupling of <b>O1</b>                               |
| <code>o1dright</code> | 1     | <code>L01dr</code> | Right-handed down-type quark coupling of <b>O1</b>                            |
| <code>o1dleft</code>  | 1     | <code>L01dl</code> | Left-handed down-type quark coupling of <b>O1</b>                             |

Internal parameters `s0fusionScalar`, `s0fusionAxial`, `o0fusionScalar`, and `o0fusionAxial` are loop-form-factor coefficients built from the corresponding top couplings,  $g_s$ ,  $v$ , and the spin-0 mass. Internal widths are `WS0`, `W00`, `WS1`, and `W01`. The tensor field **S2** has a fixed width declaration `WS2 = 2`, but no interactions are defined for it in the file.

## Physics summary

The file encodes neutral color-singlet and color-octet spin-0 and spin-1 resonances, plus a neutral singlet spin-2 state with no listed interactions. The spin-0 states couple to top quarks through scalar and pseudoscalar Yukawa-like terms and to gluons through effective  $GG$  and  $G\tilde{G}$  operators, while the spin-1 states couple chirally to Standard Model fermion currents: the singlet vector to quarks and leptons with electroweak-charge-weighted coefficients, and the octet vector to quark color currents. These interactions mediate resonance production in gluon fusion or quark annihilation and decays to  $t\bar{t}$ , dijets, charged leptons, and neutrinos depending on the state and coupling choices.

## Paper cross-check

### Lagrangian And Field Definitions

The paper defines an EFT in the **INTRODUCTION** and **SETUP** sections. The master EFT Lagrangian is Eq. (1):

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i O_i}{\Lambda^2} + \text{H.c.}$$

The field notation is given in **SETUP**:  $Q$  is the third-generation left-handed quark doublet,  $q$  is a first- or second-generation left-handed quark doublet,  $t$  is the right-handed top quark,  $u, c$  are right-handed up/charm quarks,  $\phi$  is the Higgs doublet, and  $\tilde{\phi} = i\sigma_2\phi$ .

The paper's defining dimension-six operators are Eqs. (2)-(5):

$$O_{uG}^{(1,3)} = y_t g_s (\bar{q} \sigma^{\mu\nu} T^A t) \tilde{\phi} G_{\mu\nu}^A,$$

$$O_{u\phi}^{(1,3)} = -y_t^3 (\phi^\dagger \phi) (\bar{q} t) \tilde{\phi},$$

$$O_{uG}^{(3,1)} = y_t g_s (\bar{Q} \sigma^{\mu\nu} T^A u) \tilde{\phi} G_{\mu\nu}^A,$$

$$O_{u\phi}^{(3,1)} = -y_t^3 (\phi^\dagger \phi) (\bar{Q} u) \tilde{\phi}.$$

The tree-level  $tuh$  interaction is Eq. (8), modified after removing  $u_L - t_R$  mixing to Eq. (11):

$$\mathcal{L}'_{tuh} = -C_{u\phi} \frac{m_t^2}{\Lambda^2} \frac{2y_t}{\sqrt{2}} (\bar{u} P_R t) h + \text{H.c.}$$

The paper also gives the effective  $utg$  interaction from  $O_{uG}$  in Eq. (30):

$$\mathcal{L}_{\text{Eff}} = -\frac{C_{uG}}{\Lambda^2} 2m_t g_s (\bar{u}_L \sigma^{\mu\nu} T^A t_R) \partial_\nu G_\mu^A.$$

| paper term (eq. ref)                                                                                        | reconstruction term                                                                                                | verdict                   | notes                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| SM field content<br>$Q, q, t, u, c, \phi, \tilde{\phi}$ in<br><b>SETUP</b>                                  | New fields<br>$S0, O0, S1, O1, S2$ plus<br>SM fermions                                                             | disagree                  | The paper defines only SM fields in an EFT basis; the reconstruction defines new neutral spin-0, spin-1, and spin-2 resonances absent from the paper. |
| $\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{SM}} + \sum_i C_i O_i / \Lambda^2 + \text{H.c.},$<br>Eq. (1) | Sum of explicit resonance interactions<br>$\mathcal{L}_{S0}, \mathcal{L}_{O0}, \mathcal{L}_{S1}, \mathcal{L}_{O1}$ | disagree                  | Paper is a dimension-six SM EFT; reconstruction is a simplified resonance model with independent new fields.                                          |
| $O_{uG}^{(1,3)} = y_t g_s (\bar{q} \sigma^{\mu\nu} T^A t) \tilde{\phi} G_{\mu\nu}^A,$ Eq. (2)               | None                                                                                                               | missing-in-reconstruction | No flavor-changing chromomagnetic $q_L - t_R$ operator appears.                                                                                       |
| $O_{u\phi}^{(1,3)} = -y_t^3 (\phi^\dagger \phi) (\bar{q} t) \tilde{\phi},$ Eq. (3)                          | None                                                                                                               | missing-in-reconstruction | No dimension-six flavor-changing Higgs-Yukawa operator appears.                                                                                       |
| $O_{uG}^{(3,1)} = y_t g_s (\bar{Q} \sigma^{\mu\nu} T^A u) \tilde{\phi} G_{\mu\nu}^A,$ Eq. (4)               | None                                                                                                               | missing-in-reconstruction | No opposite-flavor chromomagnetic operator involving $Q_L$ and $u_R/c_R$ appears.                                                                     |
| $O_{u\phi}^{(3,1)} = -y_t^3 (\phi^\dagger \phi) (\bar{Q} u) \tilde{\phi},$ Eq. (5)                          | None                                                                                                               | missing-in-reconstruction | No opposite-flavor Higgs-Yukawa operator involving $Q_L$ and $u_R/c_R$ appears.                                                                       |

| paper term (eq. ref)                                                                                             | reconstruction term                                    | verdict                   | notes                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hermitian conjugates of Eqs. (2)-(5), discussed after Eq. (5)                                                    | Self-conjugate resonance fields and ordinary bilinears | disagree                  | The paper requires H.c. flavor-changing operators with chirality-flipped contributions; reconstruction has real self-conjugate bosons and flavor-diagonal currents. |
| $y_t = \sqrt{2}m_t/v$ , Eq. (7)                                                                                  | Uses $m_t/v$ in scalar-top terms                       | convention                | Related normalization appears, but the reconstruction applies it to new scalar resonance couplings rather than the paper's EFT operators.                           |
| Tree-level $tuh$ interaction $-C_{u\phi}m_t^2\Lambda^{-2}(3y_t/\sqrt{2})(\bar{u}P_R t)h + \text{H.c.}$ , Eq. (8) | None                                                   | missing-in-reconstruction | Reconstruction contains no Higgs field $h$ , no $t - u/c$ flavor change, and no $P_R$ chiral structure of this kind.                                                |
| Field rotations removing $u_L - t_R$ mixing, Eqs. (9)-(10)                                                       | None                                                   | missing-in-reconstruction | Reconstruction has no flavor-changing mass mixing induced by $O_{u\phi}$ .                                                                                          |
| Rotated $tuh$ interaction $-C_{u\phi}m_t^2\Lambda^{-2}(2y_t/\sqrt{2})(\bar{u}P_R t)h + \text{H.c.}$ , Eq. (11)   | None                                                   | missing-in-reconstruction | This is the paper's physical tree-level $tuh$ vertex after diagonalization; reconstruction has no counterpart.                                                      |
| Counterterm form $O_{u\phi} \rightarrow -y_t^3(\phi^\dagger\phi - v^2/2)(\bar{q}t)\tilde{\phi}$ , Eq. (12)       | None                                                   | missing-in-reconstruction | Reconstruction has no Higgs-doublet EFT operator or subtraction of the vev-induced mixing term.                                                                     |
| Effective decomposition $\mathcal{L}_{\text{Eff}} = \mathcal{L}_{tu} + \mathcal{L}_{tuh}$ , Eqs. (25)-(27)       | None                                                   | missing-in-reconstruction | Reconstruction lacks the $u_L - t_R$ mixing and $tuh$ counterterm structure.                                                                                        |
| $utg$ dipole interaction from $O_{uG}$ , Eq. (30)                                                                | None                                                   | missing-in-reconstruction | Reconstruction has gluon field-strength operators, but they couple gluons to new scalars, not to a flavor-changing quark dipole.                                    |
| EOM-vanishing counterterm operators $O^{(1)}, O^{(2)}$ , Eqs. (33)-(34)                                          | None                                                   | missing-in-reconstruction | These are part of the paper's renormalization treatment of $O_{uG}$ ; reconstruction has no analogous EFT renormalization structure.                                |
| Renormalized effective Lagrangian with $C_{u\phi}$ , $C_{uG}$ , and mixing counterterms, Eq. (37)                | None                                                   | missing-in-reconstruction | Reconstruction does not contain Wilson coefficients $C_{u\phi}, C_{uG}$ , operator mixing, or dimension-six counterterms.                                           |

| paper term (eq. ref) | reconstruction term                                                                                                                  | verdict                 | notes                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------|
| None                 | $S0 \bar{t}t$ scalar coupling                                                                                                        | extra-in-reconstruction | No color-singlet scalar resonance $S0$ or flavor-diagonal $S0\bar{t}t$ interaction is defined in the paper.   |
| None                 | $iS0 \bar{t}\gamma^5 t$ pseudoscalar coupling                                                                                        | extra-in-reconstruction | No pseudoscalar top resonance coupling appears in the paper.                                                  |
| None                 | $S0 G_{\mu\nu}^a G^{a\mu\nu}$                                                                                                        | extra-in-reconstruction | Paper has a chromomagnetic quark-Higgs-gluon operator, not a scalar-gluon-gluon effective resonance operator. |
| None                 | $S0 G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$                                                                                                | extra-in-reconstruction | No CP-odd scalar-gluon-gluon operator is part of the paper's model.                                           |
| None                 | $O0^a \bar{t}T^a t$ scalar color-octet coupling                                                                                      | extra-in-reconstruction | Paper has no color-octet scalar resonance.                                                                    |
| None                 | $iO0^a \bar{t}\gamma^5 T^a t$ color-octet pseudoscalar coupling                                                                      | extra-in-reconstruction | Paper has no color-octet pseudoscalar top coupling.                                                           |
| None                 | $d^{abc}O0^a G_{\mu\nu}^b G^{c\mu\nu}$                                                                                               | extra-in-reconstruction | Paper has no $d^{abc}$ scalar-octet gluon-fusion operator.                                                    |
| None                 | $d^{abc}O0^a G_{\mu\nu}^b \tilde{G}^{c\mu\nu}$                                                                                       | extra-in-reconstruction | Paper has no CP-odd color-octet scalar-gluon operator.                                                        |
| None                 | $S1_\mu \bar{u}_L \gamma^\mu u_L, S1_\mu \bar{d}_L \gamma^\mu d_L, S1_\mu \bar{u}_R \gamma^\mu u_R, S1_\mu \bar{d}_R \gamma^\mu d_R$ | extra-in-reconstruction | Paper has no neutral color-singlet vector resonance or flavor-diagonal quark-current interactions.            |
| None                 | $S1_\mu \bar{e}_L \gamma^\mu e_L, S1_\mu \bar{\nu}_L \gamma^\mu \nu_L$                                                               | extra-in-reconstruction | Paper does not introduce lepton-current couplings.                                                            |
| None                 | $O1_\mu^a \bar{u}_{L,R} \gamma^\mu T^a u_{L,R}, O1_\mu^a \bar{d}_{L,R} \gamma^\mu T^a d_{L,R}$                                       | extra-in-reconstruction | Paper has no color-octet vector resonance or flavor-diagonal color-current couplings.                         |
| None                 | Tensor field $S2$ , neutral singlet, no interactions                                                                                 | extra-in-reconstruction | No spin-2 field is present in the paper.                                                                      |

## Disagreements

| item                                                                                    | severity    | what a human should check                                                                                                                  |
|-----------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Reconstruction introduces $S0, O0, S1, O1, S2$ resonance fields absent from the paper.  | substantive | Check whether <b>reconstruction.md</b> came from a different model file than the paper, possibly a generic resonance implementation.       |
| Paper's EFT master Lagrangian Eq. (1) is not reproduced.                                | substantive | Verify whether the implementation was supposed to encode the paper's dimension-six EFT or a separate phenomenological model.               |
| Missing $O_{uG}^{(1,3)}$ , Eq. (2).                                                     | substantive | Check for any hidden flavor-changing top-light-quark chromomagnetic operator in the original implementation.                               |
| Missing $O_{u\phi}^{(1,3)}$ , Eq. (3).                                                  | substantive | Check whether the implementation contains a $tuh$ FCNC Yukawa vertex under another name.                                                   |
| Missing $O_{uG}^{(3,1)}$ , Eq. (4).                                                     | substantive | Check whether chirality-flipped or conjugate flavor structures were intentionally omitted.                                                 |
| Missing $O_{u\phi}^{(3,1)}$ , Eq. (5).                                                  | substantive | Check whether the implementation supports both $t \rightarrow uh/ch$ chiralities.                                                          |
| Reconstruction lacks the paper's Hermitian-conjugate FCNC structure.                    | substantive | Check whether both decay and conjugate amplitudes are generated in the implementation.                                                     |
| Reconstruction uses $m_t/v$ normalization for unrelated scalar-top resonance couplings. | convention  | Check whether this normalization was copied from another simplified model and not from the EFT paper.                                      |
| Missing paper's tree-level $tuh$ interaction Eq. (8).                                   | substantive | Check generated Feynman rules for a $t - u/c - h$ vertex.                                                                                  |
| Missing field rotations Eqs. (9)-(10).                                                  | substantive | Check whether mass diagonalization or equivalent counterterms are implemented.                                                             |
| Missing physical rotated $tuh$ coupling Eq. (11).                                       | substantive | Check whether the implementation uses the correct factor $2y_t/\sqrt{2}$ , not the pre-rotation factor $3y_t/\sqrt{2}$ .                   |
| Missing counterterm-subtracted $O_{u\phi}$ form Eq. (12).                               | substantive | Check whether vev-induced $u_L - t_R$ mixing is removed consistently.                                                                      |
| Missing $\mathcal{L}_{tu} + \mathcal{L}_{tuh}$ counterterm structure Eqs. (25)-(27).    | substantive | Check whether NLO renormalization was implemented at all.                                                                                  |
| Missing $utg$ dipole interaction Eq. (30).                                              | substantive | Check whether the implementation contains a flavor-changing $t - u/c - g$ dipole vertex rather than only gluon-fusion resonance operators. |
| Missing EOM-vanishing counterterms Eqs. (33)-(34).                                      | substantive | Check whether the implementation is intended only for tree level or includes the NLO operator-renormalization setup.                       |

| item                                                       | severity    | what a human should check                                                                               |
|------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------|
| Missing renormalized Lagrangian Eq. (37).                  | substantive | Check whether Wilson-coefficient mixing $C_{uG} \rightarrow C_{u\phi}$ is encoded elsewhere.            |
| Extra $S0\bar{t}t$ , $S0GG$ , and $S0G\tilde{G}$ terms.    | substantive | Check whether the source implementation is for a scalar resonance model, not this FCNC Higgs EFT paper. |
| Extra $O0\bar{t}T^at$ , $O0GG$ , and $O0G\tilde{G}$ terms. | substantive | Check whether color-octet scalar resonance interactions belong to another paper or benchmark.           |
| Extra $S1$ flavor-diagonal quark and lepton currents.      | substantive | Check whether a $Z'$ -like model was accidentally reconstructed instead of the paper's EFT.             |
| Extra $O1$ color-octet vector quark currents.              | substantive | Check whether an axigluon/coloron model was mixed into the reconstruction.                              |
| Extra inert spin-2 field $S2$ .                            | substantive | Check whether the implementation file contains unused benchmark particles unrelated to the paper.       |

## Overall Assessment

The reconstruction does not match the model defined in the paper. The paper is a Standard Model EFT treatment of flavor-changing top-Higgs interactions, centered on the dimension-six operators  $O_{uG}$  and  $O_{u\phi}$ , their flavor assignments, chirality structure, Higgs-doublet dependence, and NLO QCD renormalization. The reconstruction instead describes a simplified resonance model with new scalar, vector, color-octet, and tensor fields coupled mostly to flavor-diagonal SM currents or gluon-fusion operators. Apart from incidental use of familiar SM quantities such as  $m_t/v$ ,  $g_s$ , and color generators, the field content, operator basis, charges, chirality structure, and coefficient organization are substantively different.

## Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 28 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

## Physicist sign-off

- Reviewed by: \_\_\_\_\_ Date: \_\_\_\_\_
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: